

- For the first time, a lattice-free elementary deformation unit (OSTZ) is defined.
- For the first time, dislocations arise as a collective OSTZ-chain excitation.
- For the first time, Peierls stress, core energy, and SFE are derived, not fitted.
- Parameter-free rate equation validated across 33 systems: metals, glasses, ceramics.
- For the first time, crystal and lattice-free plasticity are unified via Eshelby.